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Incidence and Risk Factors for Secondary Glaucoma in Eyes with Uveal Melanoma

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Purpose: To estimate incidence of and analyze risk factors for developing secondary glaucoma in eyes with uveal melanoma before and after diagnosis.

Design: A cross-sectional, population-based cohort study.

Participants: Seven hundred eighty-one patients (median age, 64 years; range, 14–93) consecutively diagnosed with uveal melanoma from 1997 to 2012 in a national ocular oncology service, 708 (91%) of whom received ruthenium (50%) or iodine (50%) brachytherapy.

Methods: Patient, tumor, treatment, and follow-up data were collected prospectively. Frequency and associations of melanoma-related glaucoma at tumor diagnosis were assessed. Incidence of developing secondary glaucoma after diagnosis was estimated by Kaplan–Meier analysis. Independent risk factors were modeled using Cox regression.

Main Outcome Measures: Melanoma-related glaucoma and related risk factors.

Results: Forty-five patients (5.8%; 95% confidence interval [CI], 4.2–7.6) had tumor-related secondary glaucoma at diagnosis, 34 (76%) from a narrow-to-closed angle (25 had direct angle invasion) and 10 (22%) from anterior neovascularization. Synchronous metastases were common in patients with initial secondary glaucoma (11% vs. 1.2% with incident glaucoma, $P = 0.005$). Patients with secondary glaucoma were often male (58% vs. 48% without glaucoma; $P = 0.010$) and had larger tumors (median thickness, 9.1 vs. 4.0 mm; $P < 0.001$) involving the ciliary body (43% vs. 21%; $P < 0.001$) with retinal detachment (53% vs. 30%; $P < 0.001$). One hundred and sixty-eight patients 165 of which were treated with brachytherapy developed incident tumor- or treatment-related secondary glaucoma a median of 1.7 years (range, 0.1–13.6) after tumor diagnosis. Cumulative proportion of developing secondary glaucoma was 23% (95% CI, 20–27) at 5 years. The most common mechanism was neovascularization in 119 patients (71%; 95% CI, 63–78). By multivariable regression, initial retinal detachment 3 to 4 quadrants (hazard ratio [HR], 2.18; $P < 0.001$), initial intraocular pressure 17 mmHg or higher (HR, 1.64; $P = 0.01$), and tumor thickness predicted incident secondary glaucoma.

Conclusions: Secondary glaucoma at initial uveal melanoma diagnosis predicts high risk of synchronous metastases. Although anterior neovascularization is the most common mechanism for secondary glaucoma after diagnosis, other mechanisms such as angle narrowing and anterior chamber hemorrhage are not infrequent. Initial retinal detachment and intraocular pressure with tumor thickness could inform interim assessments of intraocular pressure and neovascularization. *Ophthalmology Glaucoma 2023;6:29-41 © 2022 by the American Academy of Ophthalmology. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).*

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An untreated uveal melanoma can lead to secondary glaucoma through direct and indirect mechanisms.[1–4] The former includes invasion of the chamber angle, usually by anteriorly centered tumors, and seeding to the trabecular meshwork, as in melanomalytic glaucoma. The latter comprises secondary angle closure by anterior displacement of the iris-lens diaphragm because of swelling of the tumor or by anterior rotation of the ciliary body because of uveal effusion, and anterior segment neovascularization, mostly from retinal ischemia.

Secondary glaucoma is also a well-known complication after plaque brachytherapy, globally the most common type of radiotherapy for uveal melanoma.[5–8] Radiation-induced ischemia of the adjacent retina and, depending on the location of the tumor, the entire retina because of the disturbed circulation of the optic nerve head and of the iris all can result in anterior segment neovascularization and neovascular glaucoma.[5] However, secondary glaucoma after plaque brachytherapy is not always neovascular. Other possible mechanisms are, for example, anterior chamber angle synechia caused by angle irradiation, or angle narrowing following irradiation-induced cataract formation.

Tumor- or treatment-related secondary glaucoma is an extra burden to a patient with uveal melanoma. Besides any necessary medical therapy, which may be needed for a long period of time, it necessitates additional visits to the

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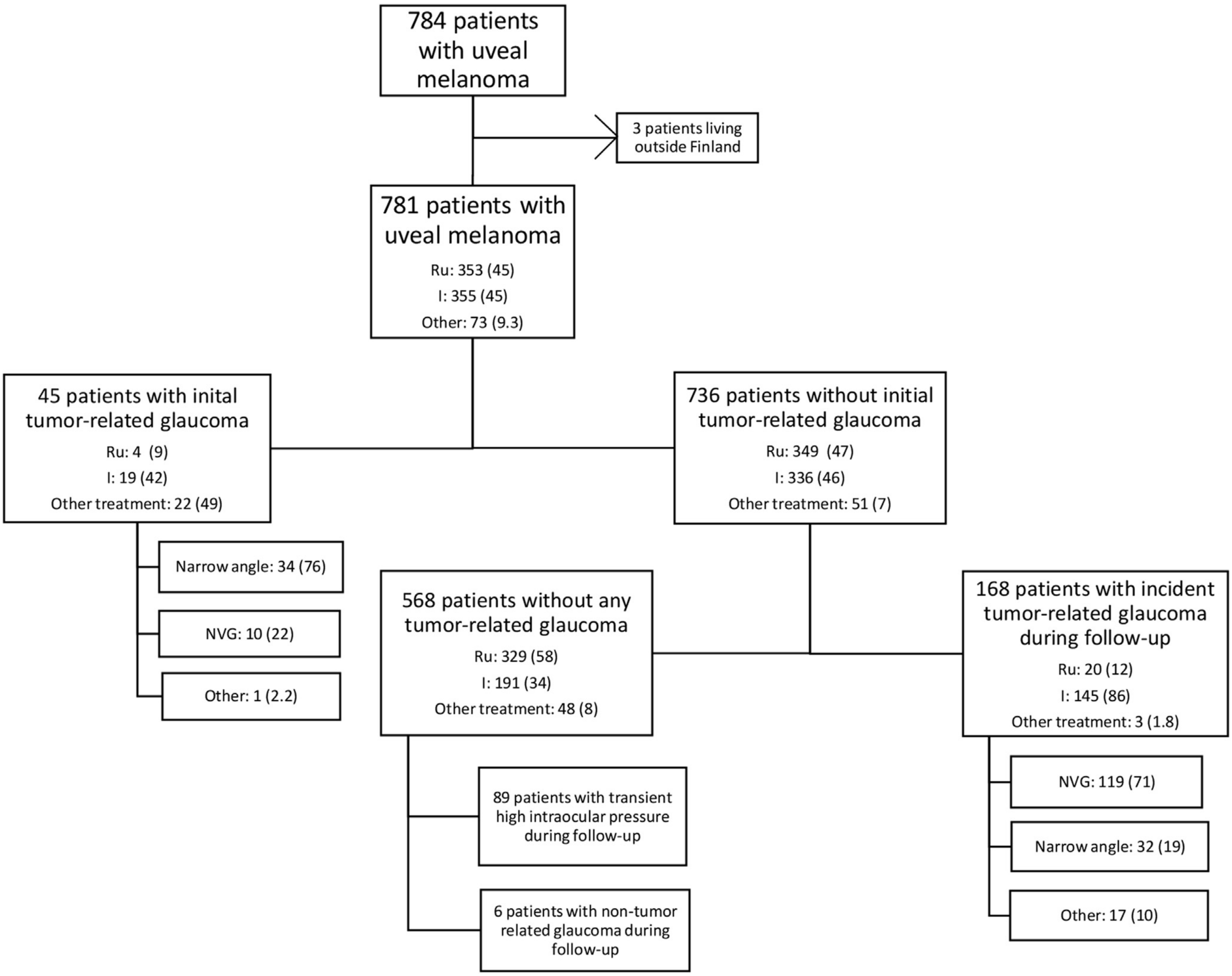

Figure 1. Patients included in the study. The number of patients (the percentage in the parentheses) treated with "Ru" (ruthenium brachytherapy), "I" (iodine brachytherapy), or "Other treatment" is listed in each subgroup. The mechanism for the secondary glaucoma is listed as "Narrow angle", "NVG" (neovascular glaucoma), or "Other."

ophthalmologist. At worst, secondary glaucoma in an eye with uveal melanoma can result in loss of any otherwise saved vision and even make the eye blind and painful. In fact, secondary glaucoma is the second most common reason for a secondary enucleation of an eye with uveal melanoma, following local tumor recurrence.[9,10]

Our aim is to assess the frequency and type of initial secondary glaucoma and to analyze the incidence and risk factors for developing secondary glaucoma after tumor diagnosis and treatment, especially with brachytherapy, in eyes with uveal melanoma in a national ocular oncology center. Previous studies on risk factors for secondary glaucoma in eyes with uveal melanoma treated with brachytherapy have been referral-based.[6–8,11–15] A better understanding of the incidence and predisposing factors should be helpful in efficiently identifying and following patients at risk and in guiding future studies in order to improve prevention, treatment, and prognosis of secondary glaucoma in eyes with uveal melanoma.

Methods

Inclusion and Exclusion Criteria

Eligible for this study were consecutive patients diagnosed with a uveal melanoma between January 1, 1997, and December 31, 2012, in the Ocular Oncology Service, Department of Ophthalmology, Helsinki University Hospital, Helsinki, Finland, the national referral center for eye cancer. Only patients residing outside Finland were excluded because of incomplete follow-up data.

This cross-sectional cohort study followed the tenets of the Declaration of Helsinki, and it was approved by the institutional review board of the Hospital Region of Helsinki and Uusimaa.

The prospective database of the Ocular Oncology Service returned 784 patients who met the inclusion criteria (Fig 1). Three patients permanently living abroad were excluded, leaving 781 patients in the analysis. Complete follow-up data were gathered until the end of 2015 by acquiring patient charts from all hospitals that had managed the uveal melanoma and secondary glaucoma, in case follow-up already had been delegated to another hospital, by permission obtained from The Finnish Institute for Health and Welfare.

Definitions

We categorized the patients in 2 groups for analysis. The *glaucoma group* consisted of patients who either had a secondary glaucoma at the time of the diagnosis of uveal melanoma or developed a secondary glaucoma during follow-up. The *reference group* consisted of patients without any tumor or its treatment-related secondary glaucoma developing.

We defined secondary glaucoma as an intraocular pressure (IOP) $\geq$ 24 mmHg together with any of the following: (1) measured on at least 3 consecutive follow-up examinations; (2) considered an indication for treatment before the third examination; or (3) associated with emergent anterior segment neovascularization. Ischemia resulting in anterior segment neovascularization could be caused by either the tumor itself or its treatment, especially with radiation; an attempt to distinguish between these 2 mechanisms was not made.

The following exceptions were applied and the patient was kept in the reference group: (1) the IOP of the fellow eye was similarly elevated and no anterior segment neovascularization was present (16 patients, IOP range 24−26 mmHg); (2) the IOP rise was related to a transient event, such as iritis or transient anterior chamber hemorrhage (3 patients, resolved without treatment), and the IOP normalized after treating the underlying condition (10 patients); (3) the IOP elevated to $\geq$ 24 mmHg postoperatively after brachytherapy but was normal without medication at 3 months, with or without temporary medical treatment in the interim, because such an increase in IOP typically is associated with transient radiation-induced swelling of a larger or anteriorly located irradiated uveal melanoma, a larger plaque in a tight orbit, or a steroid response (50 patients).

We did not exclude from the analysis patients with a nontumor related glaucoma at the time of diagnosis of uveal melanoma. In these patients, the diagnosis of secondary glaucoma during follow-up was made if the IOP increased $\geq$ 10 mmHg or an increase in glaucoma medication was needed unilaterally in the tumor eye. Anterior neovascularization together with any IOP rise and/or increase in glaucoma medication was also sufficient for the diagnosis.

Baseline and Follow-up Evaluation

Age, sex, and past ocular history were recorded. The diagnosis of uveal melanoma was based on clinical and ultrasonographic and, in selected cases, fluorescein angiography and biopsy findings. The best corrected visual acuity (BCVA) was measured using a test-type projector (Rodavist 2 and 524, Rodenstock) or display (TCP-2000A, Tomey) in even decimals from 1.0 (20/20) to 0.1 (20/200) plus 0.05 (20/400). A worse BCVA was recorded as counting fingers (CF), hand movements, light perception, and no light perception (NLP). World Health Organization (WHO) definitions for low vision (20/70 or worse) and blindness (below 20/400) were used. Loss of a vision level was recorded if confirmed at the next examination.

Intraocular pressure was measured at the time of diagnosis of the tumor and secondary glaucoma and also mostly during follow-up with the Goldmann applanation tonometer (Haag-Streit). The Icare TA01 tonometer (Icare Finland) was used for some follow-up measurements after 2003. Biomicroscopy, indirect ophthalmoscopy, fundus photography, B-scan ultrasonography, and ultrasound biomicroscopy were used to localize tumor margins, to measure tumor thickness and largest basal diameter (LBD), to estimate the distance of the posterior tumor margin to the optic disc and foveola, and to evaluate the presence of retinal detachment and vitreous hemorrhage. Gonioscopy was used to determine the anterior tumor margin, to detect neovascularization of the chamber angle, and to classify the type of glaucoma. Tumors were staged according to the eighth edition of the American Joint Committee on Cancer tumor, node, metastasis (TNM) system.[16,17]

Patients were reviewed at 1, 3, 6, and 12 months, then every 6 months until 3 to 4 years after conservative therapy, and thereafter mostly annually in the Ocular Oncology Service or in the referring university or central hospital of their place of residence. Tumor control, BCVA, IOP and glaucoma medications, lens opacities, neovascularization, retinal detachment, vitreous hemorrhage, maculopathy, optic neuropathy, local recurrence, metastases, and surgical or laser procedures were recorded.

Primary Treatment of the Tumor

The type of treatment was chosen according to tumor characteristics and patient preference. Brachytherapy, with purchased ruthenium-106 (Eckert & Ziegler BEBIG) or custom-made iodine-125 plaques, was performed in the Ocular Oncology Service. Tumor margins were localized with transillumination, using a fiberoptic probe, and with indirect ophthalmoscopy and scleral indentation. To correctly position the plaque, the extraocular muscles located in the area were disinserted as needed. Plaque position was verified intraoperatively by indirect ophthalmoscopy while indenting the plaque with a Laqua depressor or by transilluminating the plaque margins with a bent vitrectomy light probe. The plaque was sutured on the sclera. A physicist calculated the treatment time. The aim with a ruthenium-106 plaque was 100 Gy to the apex or a minimum of 250 to 350 Gy to the sclera. With an iodine-125 plaque, the aim was 80 Gy to the apex in tumors 12 mm or less thick, 70 Gy in tumors 12 to 14 mm thick, and 60 Gy in tumors more than 14 mm thick.

Local resections were performed by a senior ocular oncologist (T.T.K.). Primary enucleations were performed mostly in the Helsinki University Hospital and in part in central hospitals after the diagnosis of the uveal melanoma had been made in the Ocular Oncology Service. Transpupillary thermotherapy (TTT) was used to control focal recurrences at the tumor margin, performed with an 810-nm infrared laser (Type 426, MedArt) adapted to a slit lamp (Zeiss 30 SL-M, Zeiss). More extensive recurrences were managed either with a second brachytherapy or enucleation, depending on the extent of the recurrence, the level of remaining vision, and patient preference.

Treatment Target of Secondary Glaucoma

The primary goal of the treatment of glaucoma was to preserve vision. In eyes with little or no useful remaining vision, a secondary goal was to keep the eye painless and to preserve it for cosmetic reasons, if considered safe. Enucleation was proposed for persistently painful eyes with no useful vision and if the treated tumor could not be adequately reviewed.

Statistical Analysis

We collected follow-up data prospectively to a dedicated database (MS Access; Microsoft) and analyzed it with Stata (release 13.0, Stata Corp). Descriptive data are reported as mean with standard deviation or median with range. Fisher exact test and nonparametric test for trend was used to compare unordered and singly ordered contingency tables, respectively.[18] Kruskal−Wallis test was used to compare continuous variables. All tests were 2-tailed, and $P < 0.05$ was considered statistically significant.

The Kaplan−Meier product-limit method was used to analyze the time from the diagnosis of the primary uveal melanoma to the development of secondary glaucoma. Patients were censored if they were lost to follow-up, underwent enucleation, or died during follow-up. This method of analysis ignores competing risks and thus provides an estimate of developing incident secondary

Table 1. Characteristics of 781 Patients with Uveal Melanoma

Characteristic	All Patients, n = 781	Initial Secondary Glaucoma, n = 45	Incident Secondary Glaucoma, n = 168	*P*	Glaucoma Group, n = 213	Reference Group, n = 568	*P*
Age, median, y (range)	64 (14–93)	67 (37–87)	62 (14–85)	0.012‡	63 (14–87)	64 (14–93)	0.78‡
Age, no. (%)				0.018†			0.83†
<40	41 (5.2)	2 (4.4)	11 (6.5)		13 (6.1)	28 (4.9)	
40–49	78 (10)	3 (6.7)	17 (10)		20 (9.4)	58 (10)	
50–59	197 (25)	8 (18)	48 (29)		56 (26)	141 (25)	
60–69	209 (27)	12 (27)	42 (25)		54 (25)	155 (27)	
70–79	175 (22)	10 (22)	37 (22)		47 (22)	128 (23)	
≥80	81 (10)	10 (22)	13 (7.7)		23 (11)	58 (10)	
Sex, no. (%)				> 0.99*			0.010*
Male	395 (51)	26 (58)	98 (58)		124 (58)	271 (48)	
Female	86 (49)	19 (42)	70 (42)		89 (42)	297 (52)	
Melanocytosis, no. (%)	13 (1.7)	2 (4.4)	1 (0.6)	0.11*	3 (1.4)	1.8)	> 0.99*
Thickness, median, mm (range)	5.1 (0.2–19), n = 773	9.2 (0.8–17), n = 40	9.1 (1.7–16), n=166	0.79‡	9.1 (0.8–17), n = 206	4.0 (0.2–19), n = 567	< 0.001‡
LBD, median, mm (range)	12.4 (0.8–26), n = 763	17 (2.4–22), n = 36	15 (5.5–26), n=163	0.15‡	15.3 (2.4–26), n = 199	11.4 (0.8–25), n = 564	< 0.001‡
Distance to disc, median, mm (range)	3.0 (0–24), n = 757	7.5 (0–21), n = 31	3.6 (0–24), n=164	0.046‡	4.0 (0–24), n = 195	3 (0–22), n = 562	0.21‡
Distance to fovea, median, mm (range)	3.0 (0–24), n = 753	7.5 (0–20), n = 30	4 (0–21), n=164	0.031‡	4.5 (0–21), n = 194	3 (0–24), n = 559	< 0.001‡
Ciliary body involvement, no. (%)				0.004*			< 0.001*
Yes	211 (27)	28 (62)	63 (38)		91 (43)	120 (21)	
No	570 (73)	17 (38)	105 (63)		122 (57)	448 (79)	
Extraocular extension, no. (%)				0.58*			0.47*
Yes	21 (3)	0	4 (2.4)		4 (2)	17 (3)	
No	760 (97)	45 (100)	164 (98)		209 (98)	551 (97)	
Initial visual acuity				0.060†			< 0.001†
≥20/20	137 (18)	1 (2.2)	18 (11)		19 (9)	118 (21)	
20/25–20/40	271 (35)	14 (31)	36 (21)		50 (23)	221 (39)	
20/50–20/200	205 (26)	8 (18)	57 (34)		65 (31)	140 (25)	
CF – NLP	168 (22)	22 (49)	57 (34)		79 (37)	89 (16)	
WHO				0.078†			< 0.001†
Mild or no visual impairment	566 (72)	21 (47)	93 (55)		114 (54)	452 (80)	
Low vision	104 (13)	5 (11)	37 (22)		42 (20)	62 (11)	
Blindness	111 (14)	19 (42)	38 (23)		57 (27)	54 (9.5)	
IOP at baseline, median, mmHg (range)	15 (6–79), n=779	31 (9–79), n = 45	15 (8–25), n = 167	< 0.001‡	16 (8–79), n = 212	15 (6–31), n = 567	< 0.001‡
IOP at baseline of the fellow eye, median, mmHg (range)	16 (3–30), n=653	16 (3–27), n = 34	16 (10–30), n = 147	0.83‡	16 (3–30), n = 181	16 (8–30), n = 472	0.071‡
TNM category				N/A			N/A
T1a	218 (28)	6 (13)	54 (32)		60 (28)	158 (28)	
T1b	10 (1.3)	0	2 (1.2)		2 (0.9)	8 (1.4)	
T1d	1 (0.1)	0	0		0	1 (0.2)	
T2	5 (0.6)	0	0		0	5 (0.9)	
T2a	190 (24)	12 (27)	39 (23)		51 (24)	139 (25)	
T2b	23 (2.9)	0	4 (2.4)		4 (1.9)	19 (3.4)	
T3a	124 (16)	11 (24)	23 (14)		34 (16)	90 (16)	
T3b	80 (10)	7 (16)	18 (11)		25 (12)	55 (9.7)	
T3c	2 (0.3)	0	0		0	2 (0.4)	
T3d	2 (0.3)	0	0		0	2 (0.4)	
T4	3 (0.4)	0	0		0	3 (0.5)	

Table 1. (Continued.)

Characteristic	All Patients, n = 781	Initial Secondary Glaucoma, n = 45	Incident Secondary Glaucoma, n = 168	*P*	Glaucoma Group, n = 213	Reference Group, n = 568	*P*
T4a	39 (5.0)	5 (11)	7 (4.2)		12 (5.6)	27 (4.8)	
T4b	67 (8.6)	4 (8.9)	16 (9.5)		20 (9.4)	47 (8.3)	
T4c	1 (0.1)	0	0		0	1 (0.2)	
T4d	5 (0.6)	0	1 (0.6)		1 (0.5)	4 (0.7)	
T4e	10 (1.3)	0	3 (1.8)		3 (1.4)	7 (1.2)	
Undefined	1 (0.1)	0	1 (0.6)		1 (0.5)	0	
TNM stage, no. (%)				0.001[†,1]			< 0.001[†,1]
I	215 (28)	1 (2.2)	8 (4.8)		9 (4.2)	206 (36)	
IIA	193 (25)	1 (2.2)	27 (16)		28 (13)	165 (29)	
IIB	146 (19)	5 (11)	61 (36)		66 (31)	80 (14)	
IIIA	110 (14)	13 (29)	37 (22)		50 (23)	60 (11)	
IIIB	65 (8.3)	5 (11)	27 (16)		32 (15)	33 (5.8)	
IIIC	9 (1.2)	4 (8.9)	1 (0.6)		5 (2.4)	4 (0.7)	
IV	22 (2.8)	5 (11)	2 (1.2)	0.005*	7 (3.3)	15 (2.6)	0.63*
No stage (iris melanoma)	21 (2.7)	11 (24)	5 (3.0)	< 0.001*	16 (7.5)	5 (0.9)	< 0.001*
Retinal detachment, exudative				0.002[†,2]			< 0.001[†,2]
None noted/Around the tumor	480 (61)	26 (58)	63 (38)		89 (42)	391 (69)	
1 to 2 quadrants	166 (21)	4 (8.9)	46 (27)		50 (23)	116 (20)	
3 to 4 quadrants	114 (15)	7 (16)	55 (33)		62 (29)	52 (9.2)	
Indeterminate	21 (2.7)	8 (18)	4 (2.4)		12 (5.6)	9 (1.2)	
Vitreous hemorrhage				0.39*			< 0.001*
Present	89 (11)	6 (13)	34 (20)		40 (19)	49 (8.6)	
Primary treatment				< 0.001*[3]			< 0.001*[3]
Ruthenium brachytherapy	353 (45)	4 (8.9)	20 (12)		24 (11)	329 (58)	
Iodine brachytherapy	355 (45)	19 (42)	145 (86)		164 (77)	191 (34)	
Other treatments	73 (9.3)	22 (49)	3 (1.8)		25 (12)	48 (8)	
Observation	8 (1.0)	3 (6.7)	0		3 (1.4)	5 (8.8)	
Transpupillary thermotherapy	5 (0.6)	0	0		0	5 (8.8)	
Enucleation	50 (6.4)	16 (36)	2 (1.2)		18 (8.5)	32 (5.6)	
Exenteration	1 (0.1)	1 (2.2)	0		1 (0.47)	0	
Local resection	2 (0.2)	0	1 (0.6)		1 (0.47)	1 (4.7)	
Local resection with brachytherapy	7 (0.9)	2 (4.4)	0		2 (0.94)	5 (8.8)	

CF = counting fingers; IOP = intraocular pressure; LBD = largest basal diameter; N/A = not applicable or meaningful; NLP = no light perception; TNM = tumor, node, metastasis; WHO = World Health Organization.
*Fisher exact test, 2-tailed
†Nonparametric test for trend, 2-tailed
‡Kruskal–Wallis test, 2-tailed
[1]Tested with TNM stage I, II, and III
[2]Tested with "None noted/Around the tumor," "1 to 2 quadrants," and "3 to 4 quadrants"
[3]Tested with "Ruthenium brachytherapy," "Iodine brachytherapy," and "Other treatments"

Table 2. Initial Secondary Glaucoma at the Time of Diagnosis of Uveal Melanoma of Any Kind in 45 Patients and in 11 Patients with Iris Melanoma

Type of Glaucoma	All Uveal Melanomas, n (%)	95% CI (%)	Iris Melanomas, n (%)
Narrow angle	34 (76)	60−87	11 (100)
Tumor invading the chamber angle	25 (56)		10 (91)
Tumor compressing the chamber angle	6 (13)		
Narrow angle from total exudative retinal detachment	2 (4)		
Iris tumor with pigment dispersion and synechiae	1 (2)		1 (9)
Neovascular glaucoma	10 (22)	11−37	
Anterior chamber hemorrhage	1 (2)		

CI = confidence interval.

glaucoma in case removal of the eye and death would not prevent observing this complication.[19]

Univariable Cox proportional hazards regression was used to test if age (divided in tertiles), sex, congenital ocular melanocytosis, initial BCVA, initial IOP (tertiles), tumor thickness (tertiles), tumor LBD (tertiles), ciliary body involvement, extraocular extension, TNM stage, distance to foveola (tertiles), distance to disk (tertiles), retinal detachment, and vitreous hemorrhage were predictive of secondary glaucoma.

Multivariable Cox proportional hazards regression was used to obtain the best-fitting model to predict development of secondary glaucoma in an eye with uveal melanoma after therapy. The assumption of proportional hazards was verified by the method of Therneau and Grambsch.[20] Independent variables were considered for entry into the model when $P < 0.10$ in univariate analysis or if deemed clinically important (initial IOP). Models were compared with each other using the deviance test.

Results

The median age at the time of the diagnosis of uveal melanoma of the 781 patients was 64 years (range, 14−93); 395 were male and 386 female (Table 1). Thirty-three patients (4.2%) had a previous diagnosis of glaucoma in the tumor eye, unrelated to the melanoma: primary open-angle glaucoma (POAG) in 31 (4.0%) patients and chronic angle-closure glaucoma in 2 (0.3%) patients; 3 of the 33 had a history of glaucoma surgery in the past. Additionally, 2 patients with initial secondary glaucoma had undergone inadvertent filtering glaucoma surgery elsewhere, with failure to control IOP, before their melanoma was diagnosed. The median follow-up time was 6.2 years (range, 0 days to 18.6 years; interquartile range, 4.0−10.0 years).

Initial Secondary Glaucoma

Forty-five patients (5.8%, 95% confidence interval [CI], 4.2−7.6) had a tumor-related glaucoma at the time of diagnosis of the uveal melanoma (Fig 1). The anterior margin of the tumor extended to the ciliary body, chamber angle, or iris in 34 (76%; 95% CI, 60−87) patients, of which 11 patients had primary iris melanomas. Five (11%) of the patients had a stage IV melanoma (with synchronous metastasis).

The secondary glaucoma was caused by a narrow or closed angle in 34 (76%; 95% CI, 60−87) patients (Table 2): direct invasion of the angle by the tumor occurred in 25 (56%) patients, the tumor compressed the angle in 6 (13%) patients, a total exudative retinal detachment pushed the iris forward without neovascularization in 2 (4%) patients, and 1 eye had an iris melanoma with pigment dispersion and synechiae in the angle. Of the remaining patients, 10 (22%; 95% CI, 11−37) had neovascular glaucoma, and 1 patient had a choroidal melanoma, total exudative retinal detachment, and vitreous hemorrhage had a high IOP because of anterior chamber hemorrhage without neovascularization.

A comparison of eyes with initial vs. incident secondary glaucoma can be found in the supplemental material (Supplemental Text, available at www.ophthalmologyglaucoma.org).

Primary Treatment

Eight patients (1.0%) were observed without treating the uveal melanoma: 4 had synchronous metastases; 3 were older than 80 years and had a slowly growing tumor, 2 of which were iris melanomas; and 1 patient refused treatment. Five patients (0.6%) with a presumed incipient choroidal melanoma were treated with primary TTT.[21] Nine patients (1.2%) were managed with local resection, 7 of them with adjuvant brachytherapy. Fifty patients (6.4%) underwent primary enucleation, and 1 patient needed a primary exenteration.

Plaque brachytherapy was administered in 708 (91%) patients, 353 (50%) of whom received a ruthenium and 355 (50%) an iodine plaque. When placing the plaque, no rectus muscle was disinserted in 480 (68%) patients, 1 muscle in 197 (28%) patients, and 2 muscles in 31 (4%) patients. Accordingly, 1 anterior ciliary artery was cut with the temporal rectus muscle in 49 (7%) patients; 2 arteries with the superior, nasal, or inferior rectus muscle in 148 (21%) patients; 3 arteries with detached temporal and another rectus muscle in 13 (2%) patients; and 4 arteries with 2 detached nontemporal rectus muscles in 18 (2.5%) patients.

During follow-up, 72 (10%) and 6 (0.8%) of the 708 patients were treated with a second and third plaque, respectively, for a local recurrence.

Incident Secondary Glaucoma

Tumor- or treatment-related glaucoma developed a median of 1.7 years (range, 22 days to 13.6 years; interquartile range, 1.0−2.7 years) after primary tumor diagnosis in 168 (23%) of the 736 patients without initial secondary glaucoma (Fig 1); 165 (98%) were treated with brachytherapy. Two of the 3 remaining patients underwent primary enucleation 1.5 and 2 months after diagnosis; their glaucoma developed in the interim. The third patient had an iris melanoma that was locally resected and later treated with iodine brachytherapy for a recurrence.

Table 3. Incident Secondary Glaucoma during Follow-up in 168 Patients with a Uveal Melanoma of Any Kind and in 5 Patients with Iris Melanoma

Type of Glaucoma	All Uveal Melanomas, n (%)	95% CI (%)	Iris Melanomas, n (%)
Neovascular glaucoma	119 (71)	63−78	
Narrow angle	32 (19)	13−26	5 (100)
Chamber angle adhesions, with or without developing cataract	18 (11)		1 (20)
Tumor invading the chamber angle	7 (4)		4 (80)
Narrow angle from total exudative retinal detachment	4 (2)		
Tumor compressing the chamber angle	3 (2)		
Anterior chamber hemorrhage	8 (5)		
Open angle within irradiated volume	4 (2)		
Undetermined cause	5 (3)		

CI = confidence interval.

The most common mechanism for the incident secondary glaucoma was anterior segment neovascularization in 119 (71%; 95% CI, 63−78) patients (Table 3). The other causes were narrow angle in 32 (19%; 95% CI, 13−26) patients, persistent anterior chamber hemorrhage in 8 (5%) patients, and an open angle within the irradiated volume in 4 (2%) patients; in 5 eyes (3%), the cause remained undetermined. Five patients with primary iris melanoma developed incident secondary glaucoma.

Seven (4.2%) of the 168 patients had a previous diagnosis of glaucoma in the tumor eye, not related to their tumor. Five patients had anterior neovascularization with increased IOP level (range, 11−38 mmHg), also resulting in increase in glaucoma medication. One patient had persistent anterior chamber hemorrhage with increase in glaucoma medication, and 1 patient had a 14 mmHg increase in the IOP level resulting from the irradiation of the open angle.

Secondary Glaucoma in Patients Treated with Brachytherapy

Tumor-related glaucoma at the time of melanoma diagnosis was present in 23 (3.2%) of the 708 patients treated with ruthenium or iodine brachytherapy. An additional 165 patients (24%) developed a secondary glaucoma during follow-up, 117 (71%) of which were neovascular in type; 12% and 88% of them received primary ruthenium and iodine brachytherapy, respectively. Because the tumor was, on average, larger in the glaucoma than in the reference group, proportionally more patients who developed secondary glaucoma were treated with iodine plaques (88% vs. 39%; $P < 0.001$, Fisher exact test). The median time from tumor diagnosis to brachytherapy was 17 days (range, 1−305 days; in 2 patients, the time was more than 3 months, including a small juxtapapillary tumor first observed for 10 months, and a ring melanoma irradiated 5 months after diagnosis, controlling the IOP first).

The median follow-up time of 685 patients without initial secondary glaucoma who were treated with brachytherapy was 6.4 years (range, 55 days to 18.6 years), and the secondary glaucoma developed a median 1.7 years (range, 22 days to 13.6 years) after tumor diagnosis.

Other Types of Incident Glaucoma

Six patients developed a non−tumor-related glaucoma in the treated eye during follow-up: 4 were diagnosed with POAG (3 of them had POAG already in the fellow eye and glaucoma medication was later prescribed to the tumor eye, and 1 patient had a small T1 foveal tumor with a new diagnosis of POAG in the tumor eye). In addition, 1 patient was diagnosed with pigmentary glaucoma in the tumor eye, and 1 patient developed a persistently high IOP after vitrectomy and silicon oil tamponade for rhegmatogenic retinal detachment.

Transient high IOP, not classified as glaucoma, was seen in 89 patients (Table S1, available at www.ophthalmologyglaucoma.org/).

Glaucoma Group vs. Reference Group

The median age at the time of diagnosis of the primary uveal melanoma (Table 1) in the glaucoma group (63 years; range 14−87) was comparable with that in the reference group without secondary glaucoma (64 years; range 14−93). In the glaucoma group, proportionally more patients were male (58% vs. 48%; $P = 0.010$, Fisher exact test), the tumor was larger (median thickness, 9.1 vs. 4.0 mm; LBD 15.3 vs. 11.4 mm; $P < 0.001$ for both, Kruskal−Wallis test), and ciliary body involvement was more common (43% vs. 21%, $P < 0.001$, Fisher exact test). A difference in TNM stages was consequently observed ($P < 0.001$, nonparametric test for trend). The initial visual acuity was worse in the glaucoma group (low vision, 20% vs. 11%; blindness, 27% vs. 9.5%; $P < 0.001$, nonparametric test for trend).

Retinal detachment and vitreous hemorrhage were more common in the glaucoma group (53% vs. 30%, and 19% vs. 8.6%, respectively; $P < 0.001$ for both, Fisher exact test).

Other Treatments During Follow-up

Transpupillary thermotherapy was used as an adjuvant treatment in 7 patients and to control a local tumor recurrence at the tumor margin in 16 patients. Twelve patients received TTT to manage radiation retinopathy, exudative reaction, or recurrent tumor hemorrhage. Details of surgeries not directly involving the tumor are listed in the supplement (Supplemental Text, available at www.ophthalmologyglaucoma.org/).

Forty-nine eyes (6%) were secondarily enucleated, 32 eyes because of a tumor recurrence, and 16 eyes for being blind and painful. Two eyes were exenterated because of an extraocular recurrence. One patient with a dislocated intraocular lens compromising visibility underwent enucleation instead of scleral fixation and silicon oil removal, according to patient preference.

Table 4. Univariate Cox Proportional Hazards Regression of Developing Incident Glaucoma

Variable	Coefficient (SE)	Wald Chi-Square	P	Hazard Ratio (95% CI)
Age, y				
Tertile 1 (13.9−57.6)	Reference			1.00
Tertile 2 (57.7−69.8)	−0.247 (0.186)	1.77	0.18	0.78 (0.54−1.12)
Tertile 3 (69.9−93.2)	−0.112 (0.188)	0.36	0.55	0.89 (0.62−1.29)
Sex				
Female	Reference			1.00
Male	0.391 (0.157)	6.25	0.012	1.48 (1.09−2.01)
Melanocytosis				
Present	−0.984 (1.003)	0.96	0.33	0.37 (0.05−2.67)
Ciliary body involvement				
Present	0.914 (0.161)	32.38	< 0.001	2.49 (1.82−3.42)
Extraocular extension				
Present	−0.272 (0.506)	0.29	0.59	0.76 (0.28−2.05)
Vitreous hemorrhage				
Present	0.946 (0.193)	24.11	< 0.001	2.58 (1.77−3.76)
Retinal detachment				
None noted/Around the tumor	Reference			1.00
1 to 2 quadrants	0.872 (0.194)	20.16	< 0.001	2.39 (1.63−3.50)
3 to 4 quadrants	1.767 (0.186)	90.44	< 0.001	5.85 (4.07−8.42)
LBD				
Tertile 1 (0.8−10.3 mm)	Reference			1.00
Tertile 2 (10.4−14.6 mm)	0.867 (0.244)	12.60	< 0.001	2.38 (1.47−3.84)
Tertile 3 (14.7−26 mm)	1.811 (0.229)	62.57	< 0.001	6.11 (3.90−9.58)
Distance to disc				
Tertile 1 (0−1.7 mm)	Reference			1.00
Tertile 2 (1.8−5.3 mm)	−0.203 (0.195)	1.10	0.30	0.82 (0.56−1.19)
Tertile 3 (5.5−24 mm)	0.179 (0.188)	0.90	0.34	1.20 (0.83−1.73)
Distance to fovea				
Tertile 1 (0−1.5 mm)	Reference			1.00
Tertile 2 (1.6−5.3 mm)	0.285 (0.201)	2.02	0.16	1.33 (0.90−1.97)
Tertile 3 (5.5−24 mm)	0.818 (0.197)	9.80	0.002	1.85 (1.26−2.72)
IOP				
Tertile 1 (6−14 mmHg)	Reference			1.00
Tertile 2 (15−16 mmHg)	0.050 (0.187)	0.067	0.79	1.05 (0.73−1.52)
Tertile 3 (17−31 mmHg)	−0.004 (0.187)	0.0004	0.98	1.00 (0.69−1.44)
Visual acuity				
>20/20	Reference			1.00
20/25−20/40	0.073 (0.289)	0.063	0.80	1.08 (0.61−1.89)
20/50−20 /200	0.967 (0.271)	12.74	< 0.001	2.63 (1.55−4.47)
CF−NLP	1.543 (0.271)	32.38	< 0.001	4.68 (2.75−7.97)

CF = counting fingers; CI = confidence interval; IOP = intraocular pressure; LBD = largest basal diameter; NLP = no light perception; SE = standard error.

One blind, painful eye in a patient with a poor general condition was eviscerated.

Factors Predictive of Incident Secondary Glaucoma

Excluding the 45 patients with initial secondary glaucoma and 1 with no follow-up data, 735 patients remained (median follow-up, 6.2 years; range, 0.1−18.6; interquartile range, 4.0−10.0).

By Kaplan−Meier analysis and univariable, single-failure, Cox proportional hazard regression analysis (Table 4), the cumulative proportion of developing secondary glaucoma was 23% (95% CI, 20−27) at 5 years and in patients treated with brachytherapy, 24% (95% CI, 21−28). Independent variables associated with incident secondary glaucoma were male sex (Fig 2A; hazard ratio [HR], 1.48; $P = 0.012$), ciliary body involvement (Fig 2B; HR, 2.49; $P < 0.001$), LBD (Fig 2C; tertile 1 vs. 2 and tertile 1 vs. 3; HR, 2.38 and 6.11, respectively; both $P < 0.001$), initial retinal detachment (Fig 2D; none vs. 1−2 and none vs. 3−4 quadrants; HR, 2.39 and 5.85, respectively; both $P < 0.001$), initial vitreous hemorrhage (Fig 2E; HR, 2.58; $P < 0.001$), initial visual acuity (Fig 2F; 20/25−20/40 vs. 20/50−20/200 and 20/25−20/40 vs. CF−NLP; HR, 2.63 and 4.68, respectively; both $P < 0.001$), and distance to foveola (Fig S1A, available at www.ophthalmologyglaucoma.org; tertile 1 vs. 2 and tertile 1 vs. 3; HR, 1.33 and 1.85; $P = 0.16$ and $P = 0.002$, respectively). Tumor thickness (Fig 2G) and TNM stage (Fig 2H) did not meet the proportional hazards assumption. Visual acuity was omitted from further analysis as being dependent on most other aforementioned variables. Kaplan−Meier plot of time to the development of secondary glaucoma during follow-up according to primary treatment with ruthenium or iodine brachytherapy and number of arteries cut during the plaque insertion in the primary

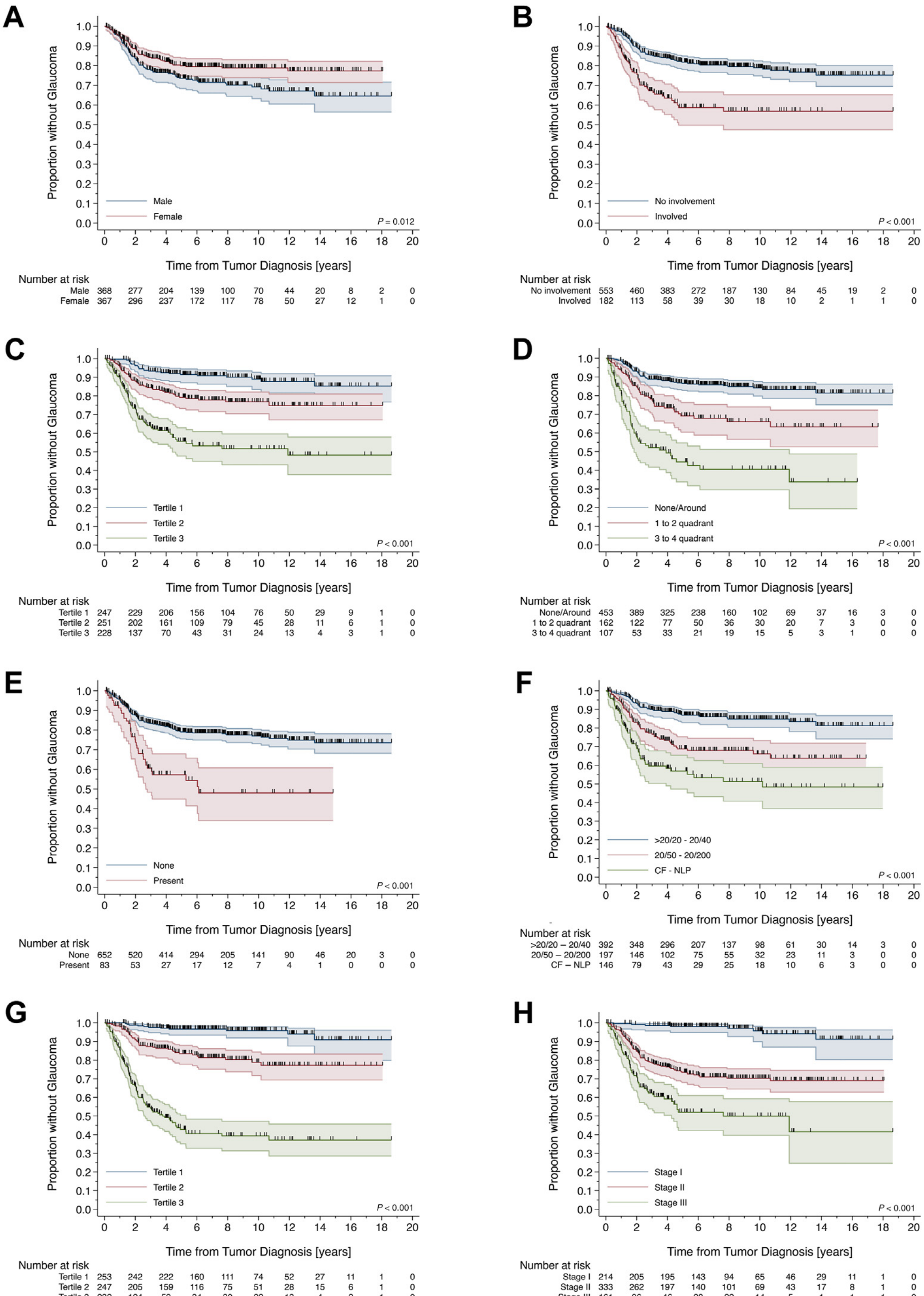

Figure 2. Kaplan–Meier plot of time to the development of secondary glaucoma during follow-up according to (**A**) sex, (**B**) ciliary body involvement, (**C**) largest basal diameter divided in tertiles, (**D**) retinal detachment, (**E**) vitreous hemorrhage, (**F**) visual acuity, (**G**) tumor thickness divided in tertiles, and (**H**) TNM stage I, II, and III. CF = counting fingers; NLP = no light perception.

treatment are shown in Figs S1B and S1C (available at www.ophthalmologyglaucoma.org).

In bivariable analysis, retinal detachment as a known risk factor for developing secondary glaucoma was combined in turn with the other univariable risk factors identified (Table S2, available at www.ophthalmologyglaucoma.org). Ciliary body involvement, LBD, distance to disc and foveola, and vitreous hemorrhage retained significance. The model with retinal detachment and LBD was the best model by the deviance test and was chosen for further modeling. Ciliary body involvement, distance to foveola, vitreous hemorrhage, and initial IOP retained significance in trivariable models (Table S3, available at www.ophthalmologyglaucoma.org). The best-fitting model was the one that included distance to foveola. In quadrivariable

Table 5. Cox Proportional Hazards Regression of Developing Incident Glaucoma, Stratified by Tumor Thickness

Variable	Coefficient (SE)	Wald Chi-Square	P	Hazard Ratio (95% CI)
Model 1: -2 Log likelihood = 1548.77				
Retinal detachment				
None noted/Around the tumor	Reference			1.00
1 to 2 quadrants	0.215 (0.210)	1.04	0.31	1.24 (0.82–1.87)
3 to 4 quadrants	0.862 (0.224)	14.82	< 0.001	2.37 (1.53–3.67)
LBD				
Tertile 1 (0.8–10.3 mm)	Reference			1.00
Tertile 2 (10.4–14.6 mm)	−0.234 (0.279)	0.71	0.40	0.79 (0.46–1.37)
Tertile 3 (14.7–26 mm)	−0.035 (0.286)	0.01	0.90	0.97 (0.55–1.69)
Distance to fovea				
Tertile 1 (0–1.5 mm)	Reference			1.00
Tertile 2 (1.6–5.3 mm)	−0.13 (0.205)	0.004	0.95	0.99 (0.66–1.47)
Tertile 3 (5.5–24 mm)	0.243 (0.215)	1.28	0.26	1.27 (0.84–1.94)
IOP				
Tertile 1 (6–14 mmHg)	Reference			1.00
Tertile 2 (15–16 mmHg)	0.210 (0.197)	1.14	0.28	1.23 (0.84–1.81)
Tertile 3 (17–31 mmHg)	0.506 (0.196)	6.66	0.01	1.66 (1.13–2.44)
Model 2: -2 Log likelihood = 1575.37				
Retinal detachment				
None noted/Around the tumor				1.00
1 to 2 quadrants	0.162 (0.205)	0.62	0.43	1.18 (0.79–1.76)
3 to 4 quadrants	0.779 (0.203)	14.67	< 0.001	2.18 (1.46–3.25)
IOP				
Tertile 1 (6–14 mmHg)				1.00
Tertile 2 (15–16 mmHg)	0.199 (0.193)	1.06	0.30	1.22 (0.84–1.78)
Tertile 3 (17–31 mmHg)	0.496 (0.194)	6.55	0.01	1.64 (1.12–2.40)

CI = confidence interval; IOP = intraocular pressure; LBD = largest basal diameter; SE = standard error.

analysis, the best-fitting model additionally included initial IOP (Table S4, available at www.ophthalmologyglaucoma.org). Further modeling was not feasible because the number of patients in many subgroups was too small. No significant interactions were detected in the final model (data not shown).

We then considered stratifying the model by tumor thickness, chosen over TNM stage because the latter is, in part, based on LBD already in the model. In this model, LBD and distance to foveola lost significance (Table 5, Model 1). The alternative final model thus consists of initial retinal detachment (HR, 2.18 for 3–4 quadrants; $P < 0.001$) and initial IOP (HR, 1.64 for 17 mmHg or higher; $P = 0.01$), stratified by tumor thickness (Table 5, Model 2).

Finally, we considered a model in which initial IOP was divided in 2, normal ($\leq$ 21 mmHg) and elevated ($>$ 21 mmHg). In this model, the initial IOP lost significance.

Discussion

In our population-based national cohort of 781 consecutive patients with a primary uveal melanoma of any location, 5.8% had tumor-related glaucoma at the time of primary tumor diagnosis. This percentage is twice as high than reported in a large referral-based series of 2111 patients, 3% by the same criterion for IOP.[3] In that study, some patients with tumor-related glaucoma might have undergone enucleation locally and might not have been referred.

The cumulative proportion of incident secondary glaucoma after treatment during a median follow-up of 6.4 years was high, 23% at 5 years, and 98% of the patients who developed secondary glaucoma in this series were treated with brachytherapy. Variable definitions of secondary glaucoma, indications for brachytherapy, characteristics of primary tumors based on referral, and follow-up practices make it challenging to compare the frequencies of secondary glaucoma in an informed way. No other studies have reported the global frequency of secondary glaucoma irrespective of the primary tumor treatment modality. Our estimate of 24% in patients treated with brachytherapy is higher than in several earlier studies[6–8,10–15,22–33] (Table S5, available at www.ophthalmologyglaucoma.org) although we used a higher IOP criterion than some of them. This might reflect the relatively long follow-up period in our study (when comparing results reported "at last follow-up") and the policy of offering conservative therapy as an option even when the melanoma was large.[34] Although tumor-related glaucoma developed a median of 1.7 years after diagnosis of the tumor, the longest interval was over 13 years. Moreover, in many previous studies, only neovascular glaucoma was reported. Retinal ischemia and anterior segment neovascularization are known complications of plaque brachytherapy, and almost three quarters (71%) of the eyes with incident secondary glaucoma indeed had neovascular glaucoma. However, almost one third (29%) developed other types of glaucoma, most often through secondary narrow angle mechanism.

In two thirds of patients with high IOP at the time of tumor diagnosis, the mechanism was anterior location of the melanoma, especially in the iris, with direct angle invasion or angle compression. An important observation was that synchronous metastases, which are seen in 2% to 3% of all patients with uveal melanoma, were 4 times more common in the initial secondary glaucoma subgroup (11%) compared with the incident secondary glaucoma subgroup. Tumors with synchronous metastases are usually large and the diagnosis thus may be late, allowing gradual painless elevation of the IOP through variable mechanisms.[35] Moreover, angle invasion, leading to secondary glaucoma, is a known risk factor for developing metastasis from iris melanoma in general.[36–38] The initial visual acuity in the glaucoma group also was worse; about one quarter of the eyes (27%) were blind by World Health Organization criteria, with larger tumors and higher frequency of retinal detachment and vitreous hemorrhage.

It is challenging to detect glaucomatous optic neuropathy in eyes with uveal melanoma, because the tumor and its treatment lead to confounding visual field defects and potentially to radiation optic neuropathy. The definition of tumor-related glaucoma has varied but is most often based on the IOP in the tumor eye with a cutoff, if given, from 20 to 24 mmHg.[3,6–8,10,12,13] It is straightforward to find eyes with a specified IOP; however, this definition also includes patients with transient high IOP without true secondary glaucoma. Some authors have addressed this, e.g., by requiring more than one measurement to confirm an elevated IOP,[6,7] a normal IOP in the fellow eye,[11] or need for IOP lowering treatment.[6,8] We used the threshold $\geq$ 24 mmHg in part because the higher the IOP, the more reliably it identifies eyes that would develop glaucomatous damage. We considered additionally the requirement of a persistently high IOP, need for medication, the IOP of the fellow eye, and the likely cause of the high IOP. A significant number, 11% of eyes treated conservatively, experienced transiently high IOP. We did not exclude patients with a previous diagnosis of a non–tumor-related glaucoma, 4% of all patients, but a criterion appropriate for tumor-related glaucoma was used in these patients.

We found, not unexpectedly, by multivariable analysis that tumor thickness, retinal detachment, and initial IOP were independent risk factors for developing incident tumor-related glaucoma in eyes with uveal melanoma. Increasing tumor size, especially increasing thickness that results in higher doses of irradiation to surrounding structures, is a well-established risk factor for anterior segment neovascularization and neovascular glaucoma after brachytherapy.[6,8,12,13,15,39] Retinal detachment is a well-known contributor to long-term retinal ischemia that contributes to their development.[40] The extent of exudative retinal detachment is furthermore related to larger tumor size and posterior location, reflected in tumor distance to foveola in our multivariable analysis.[41] The latter also increases radiation dose to the optic nerve head, contributing to ischemia. Our study also confirms the earlier findings that the initial IOP is one factor predicting incident secondary glaucoma during follow-up, possibly reflecting the aqueous outflow capacity.[6,7] In our study, already an initial IOP level of 17 mmHg or higher (as defined by being the 3rd tertile) predicted incident secondary glaucoma. An association was not statistically demonstrable if the IOP was divided by the upper normal of 21 mmHg, suggesting that higher values in the normal range also are predictive. Although all these factors are known, they have, to the best of our knowledge, not been estimated in a population-based setting or considering all treated eyes.

Of other risk factors, age, sex, iris location, ciliary body involvement, and tumor distance to disc, which have previously been proposed as risk factors for the development of secondary glaucoma after brachytherapy,[7,11,14] none were independently associated with secondary glaucoma in our multivariable analysis. Age and sex are subject to variable bias from the competing cause of death because the risk of death differs by sex and age. When patients who die before experiencing the event of interest are censored, the assumption of being at equal risk of the event of interest both before and after censoring is violated, thus resulting in the bias. This inflates all subsequent time-to-event estimates. All assessed variables are subject to the competing risk of secondary enucleation, which, in our series, was relatively infrequent.[8,12,15]

It is reassuring that 568 patients (73%) did not develop any tumor- or treatment-related glaucoma during a median follow-up of 6.2 years. Our findings further inform clinical practice to increase the proportion of eyes that avoid incident secondary glaucoma developing. Secondary glaucoma can arise long after primary treatment, at a time when review for tumor control already takes place once a year. Our data suggest that more frequent follow-up might still be reasonable in patients with thick tumors, initial retinal detachment, and initial higher IOP. In the future, prophylactic antivascular endothelial growth factor injections targeted to patients at risk could also be evaluated. This study suggests that anteriorly located tumors increase IOP early, reinforcing the importance of considering uveal melanoma in eyes with unilateral high IOP of unknown etiology.[42,43] The iris and the anterior chamber angle should be examined carefully with gonioscopy for invasion and neovascularization related to a more posterior tumor, but other mechanisms for the high IOP are also common.

In conclusion, up to 29% of patients in a population-based setting with a primary uveal melanoma developed tumor- or treatment-related secondary glaucoma, either by the time of tumor diagnosis (6%) or during follow-up (23% at 5 years), reflecting in our series in part our policy of offering brachytherapy as one option also for large tumors. The most common mechanism for elevated IOP was an anteriorly located tumor blocking the chamber angle (69%) at the time of tumor diagnosis and neovascular glaucoma (71%) following treatment of the tumor. However, other mechanisms were relatively frequent. Tumor thickness, initial retinal detachment, and initial IOP as independent risk factors for incident secondary glaucoma after treatment may serve as a guide to review the patient for emerging glaucoma more frequently than they would be reviewed for tumor control.

Footnotes and Disclosures

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Analysis and interpretation: Stadigh, Puska, Kivelä

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Abbreviations and Acronyms:

BCVA = best corrected visual acuity; **CF** = counting fingers; **CI** = confidence interval; **HR** = hazard ratio; **IOP** = intraocular pressure; **LBD** = largest basal diameter; **N/A** = not applicable or meaningful; **NLP** = no light perception; **POAG** = primary open-angle glaucoma; **SE** = standard error; **TNM** = tumor node, metastasis; **TTT** = transpupillary thermotherapy; **WHO** = World Health Organization.

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